LECTURE PLAN

Subject: Digital Electronics (NECE102), Class: I Semester, B. Tech (Common) Winter Semester (2025-26)

Sl. No.	Topics	Day (tentative)
1.	Introduction to Number Systems and Conversion Techniques	Day 1
2.	Continue…..	Day 2
3.	1’s and 2’s complement	Day 3
4.	Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR)	Day 4
5.	Boolean Algebra: Laws and Theorems	Day 5
6.	Boolean Function Implementation Using NAND/NOR Gates	Day 6
7.	Determination of Boolean Functions from Digital Circuits	Day 7
8.	Continue…..	Day 8
9.	SOP Standard Forms of Logical Functions	Day 9
10.	POS Standard Forms of Logical Functions	Day 10
11.	Simplification of Logical Functions Using Boolean Algebra	Day 11
12.	Introduction to Karnaugh Maps (K-Maps)	Day 12
13.	K-Map Simplification (2-variable Functions)	Day 13
14.	K-Map Simplification (3-variable Functions)	Day 14
15.	K-Map Simplification (4-variable Functions)	Day 15
16.	Don't Care Conditions in K-Map Simplification	Day 16
17.	Introduction to Combinational Logic Circuits	Day 17
18.	Design of Adders (Half Adder, Full Adder)	Day 18
19.	Design of Subtractors (Half Subtractor, Full Subtractor)	Day 19
20.	Magnitude Comparator Design	Day 20
21.	Code Converters (Binary to Gray, Gray to Binary, BCD to Excess-3)	Day 21
22.	Multiplexers: Design and Applications	Day 22
23.	Demultiplexers: Design and Applications	Day 23
24.	Decoders and Encoders	Day 24
25.	Introduction to Sequential Circuits	Day 25
26.	Latches: SR Latch, D Latch	Day 26
27.	Flip-Flops: SR, JK Flip-Flops	Day 27
28.	Flip-Flops: D and T Flip-Flops	Day 28
29.	Asynchronous Counters: Design and Operation	Day 29
30.	Continue…..	Day 30
31.	Synchronous Counters: Design and Operation	Day 31
32.	Continue…..	Day 32
33.	Shift Registers: Types and Applications	Day 33
34.	Continue…..	Day 34
35.	Introduction to State Tables and State Diagrams	Day 35
36.	Analysis of State Tables and State Diagrams	Day 36
37.	Timing Circuits: Concepts and Examples	Day 37
38.	Timing Circuit Design and Implementation	Day 38
39.	Semiconductor Memories: RAM and ROM	Day 39
40.	Programmable Logic Devices: PAL and PLA	Day 40
41.	Introduction to Microprocessors	Day 41
42.	Continue…..	Day 42

Textbook:

1. Digital Logic and Computer Design by M. Morris Mano, Prentice-Hall, 2015.
2. Floyd T.L, Digital Fundamentals, 10/e, Pearson Education, 2011.

Reference Books:

1. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.
2. Digital Logic and Computer Design by M. Morris Mano, Prentice-Hall, 2015.
3. Digital Principles and Applications by Malvino & Leach, 8/e McGraw Hill, 2014.
4. Modern Digital Electronics by R P Jain, 4/e McGraw-Hill Education, 2009.
5. Digital Integrated Electronics by Taub & Schilling, McGraw Hill, 2008.

Academic calendar 2025-26

https://people.iitism.ac.in/~academics/assets/academic_files/AC%202025-26%20V12.pdf

To ensure 14 weeks of classes:

- **Thursday, 15.01.2026** will be working with **Friday timetable** All students except Exec. MBA and Exec. M. Tech..
- **Wednesday, 18.03.2026** will be working with **Friday timetable** All students except Exec. MBA and Exec. M. Tech..
- **Thursday, 16.04.2026** will be working with **Tuesday timetable** All students except Exec. MBA and Exec. M. Tech.

No Class Day	Holidays
• **16.01.2026 on account of SRIJAN**	• **26.01.2026 on account of Republic Day** • **04.03.2026 on account of Holi** ***(Tentative)*** • **20.03.2026 on account of Idu'l Fitr** (subject to visibility of moon) ***(Tentative)*** • **31.03.2026 on account of Mahavir Jayanti** ***(Tentative)*** • **03.04.2026 on account of Good Friday** ***(Tentative)*** • **14.04.2026 on account of Ambedakar Jayanti** ***(Tentative)*** • **01.05.2026 on account of Buddha Purnima** ***(Tentative)***